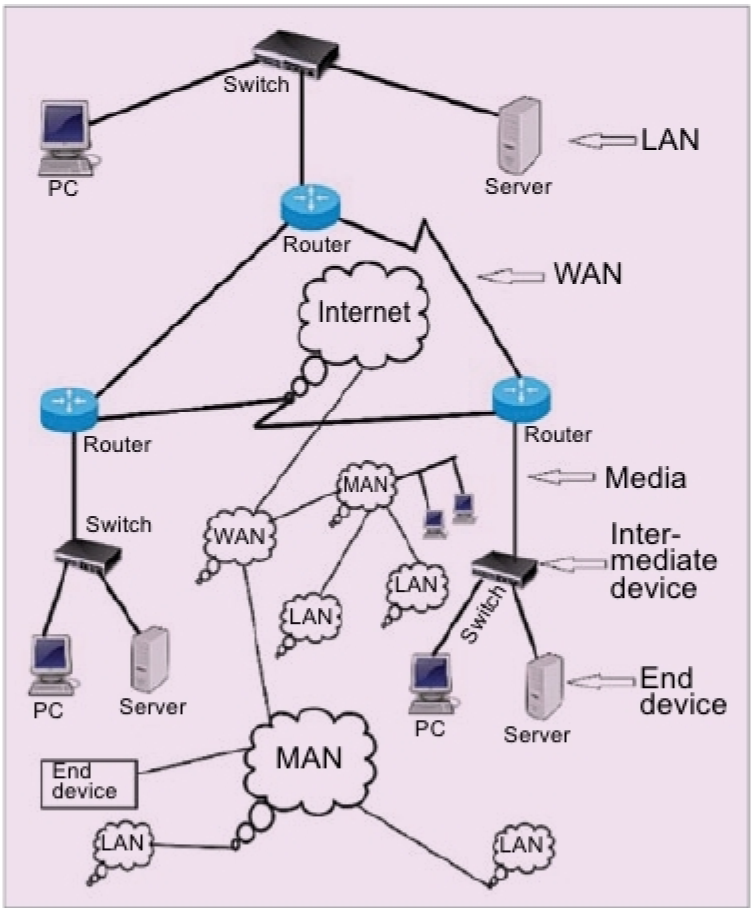

Fig. 1: The Internet, a network of networks

respectively. One resolution to fix the incompatibilities of communication between all devices connected in these networks, when interconnected, can be using several protocol converters, like A to B and B to A for communication between networks a and b; B to C and C to B for communication between networks b and c; and A to C and C to A for communication between networks a and c. Altogether, as many as six different protocol converters are needed.

So, what will happen when thousands of computer networks all over the world are connected? Cost-wise, it will be huge, and administrative jurisdiction over the converters will remain debatable.

The Internet, thereafter

The Internet (Fig. 1) came as a solution for world-wide systems of interconnected computer networks to take care of the above-mentioned challenge of global connection. It does so by using a single global protocol known as Internet Protocol (IP) over either Transmission Control Protocol (TCP) or Universal Datalink Protocol (UDP).

TCP/IP as well as UDP/IP are known as protocol suites. The Internet is best defined as a global network to link globally-distributed computers to share, exchange and provide services of World Wide Web (WWW), e-mail, voice, video, messaging and multimedia services.

The Internet, or simply the net, may also be called a network of networks. It connects millions of local area networks (LANs), metropolitan area networks (MANs), wide area networks (WANs), academic, business and state networks, which are geographically distributed all over the globe.

The Internet is the world's single backbone network that connects other networks and computers using an open system of architecture of TCP/IP or UDP/IP. For accessing the Internet, users are required to take the service of a local Internet service provider (ISP). An ISP acts as a middleman between users and the Internet.

The concept of the Internet originated in Advanced Research Projects Agency (ARPA) in the US. It was first implemented in December 1990 in CERN, Switzerland. Growth of the Internet is more than the growth rate of any other technology. A glimpse of its leaps and bounds growth is given below.

- 44 million Internet users in 1995
- 413 million Internet users in 2000
- 3.7 billion Internet users in 2016 (49.5 per cent of world population)
- 4.2 billion Internet users in 2017 (54.4 per cent of world population)

The first Asian to win a Nobel Prize is an Indian. He was Rabindra Nath Tagore, the great poet. His established far-reaching thinking was of a one world, one village. The great idea of a genius son of India is, in fact, on the verge of being achieved through the Internet.

TCP/IP and UDP/IP protocol suites

A protocol defines a set of rules for the functional operation of any computing system. As part of the process, a protocol resolves the problems of incompatibility and the overlaps existing in different systems. A protocol is also followed in offices. For example, a per-